

Relations and Orderings

Math Foundations — Concept 03

1 Relations on a Set

A *binary relation* on a set A is a subset $R \subseteq A \times A$. We write $a R b$ as shorthand for $(a, b) \in R$.

First, an example

Before any abstract axioms, look at congruence mod 3 on $A = \{0, 1, 2, 3, 4, 5, 6, 7, 8\}$. Declare $x \equiv y \pmod{3}$ when $3 \mid (x - y)$. The equivalence classes are exactly the three residue buckets:

- $[0] = \{0, 3, 6\}$
- $[1] = \{1, 4, 7\}$
- $[2] = \{2, 5, 8\}$

Notice three features: every element of A lands in exactly one bucket (covering + disjoint $\Rightarrow$ a *partition*); the buckets are determined by the relation, not chosen by hand; and the relation $\equiv \pmod{3}$ is reflexive ($x - x = 0$), symmetric ($3 \mid (x - y) \Leftrightarrow 3 \mid (y - x)$), and transitive. The abstract definitions below distil exactly these three properties.

Definition 1 (Reflexivity). A relation R on A is *reflexive* if

$$\forall a \in A, \quad a R a.$$

Read aloud: “for all a in A , a is related to a .”

Definition 2 (Symmetry). A relation R on A is *symmetric* if

$$\forall a, b \in A, \quad a R b \implies b R a.$$

Read aloud: “for all a and b in A , if a is related to b then b is related to a .”

Definition 3 (Transitivity). A relation R on A is *transitive* if

$$\forall a, b, c \in A, \quad (a R b \wedge b R c) \implies a R c.$$

Read aloud: “for all a, b, c in A , if a is related to b and b is related to c , then a is related to c .”

Definition 4 (Antisymmetry). A relation R on A is *antisymmetric* if

$$\forall a, b \in A, \quad (a R b \wedge b R a) \implies a = b.$$

Read aloud: “for all a and b in A , if a is related to b and b is related to a , then a equals b .”

Definition 5 (Equivalence relation). A relation $\sim$ on A is an *equivalence relation* if it is reflexive, symmetric, and transitive. For each $a \in A$ the *equivalence class* of a is

$$[a] = \{x \in A : a \sim x\},$$

Read aloud: “the class of a equals the set of all x in A such that a is equivalent to x .” and the *quotient set* is $A/\sim = \{[a] : a \in A\}$.

Definition 6 (Partition). A *partition* of A is a family $\mathcal{P} \subseteq \mathcal{P}(A)$ of nonempty subsets of A such that

(P1) $\bigcup_{B \in \mathcal{P}} B = A$ (covering);

(P2) $B, B' \in \mathcal{P}$ with $B \neq B' \implies B \cap B' = \emptyset$ (pairwise disjoint).

Definition 7 (Partial order). A relation $\leq$ on A is a *partial order* if it is reflexive, antisymmetric, and transitive. It is a *total order* if additionally $\forall a, b \in A, a \leq b \vee b \leq a$.

2 Equivalence Relations Correspond to Partitions

We now prove the central structural theorem of this concept.

Lemma 1. *Let $\sim$ be an equivalence relation on A , and let $a, b \in A$. Then $[a] = [b]$ if $a \sim b$, and $[a] \cap [b] = \emptyset$ otherwise.*

Proof. Suppose $a \sim b$. If $x \in [a]$, then $a \sim x$, so by symmetry $x \sim a$, and combined with $a \sim b$ transitivity gives $x \sim b$, hence $b \sim x$ by symmetry, i.e. $x \in [b]$. The reverse inclusion is symmetric, so $[a] = [b]$.

Suppose instead $a \not\sim b$, but for contradiction take $x \in [a] \cap [b]$. Then $a \sim x$ and $b \sim x$. By symmetry $x \sim b$, and transitivity yields $a \sim b$, contradicting the hypothesis. Hence $[a] \cap [b] = \emptyset$. $\square$

Theorem 2 (Equivalence relations $\leftrightarrow$ partitions). *Let A be a set. There is a bijection between the set of equivalence relations on A and the set of partitions of A , given by*

$$\Phi(\sim) = \{[a]_\sim : a \in A\}, \quad \Psi(\mathcal{P}) = \{(a, b) \in A \times A : \exists B \in \mathcal{P}, a, b \in B\}.$$

Read aloud: “ Φ of $\sim$ is the set of equivalence classes $[a]$ for a in A ; Ψ of $\mathcal{P}$ is the set of pairs (a, b) in $A \times A$ such that there exists a block B in $\mathcal{P}$ containing both a and b .”

Proof. Step 1: $\Phi(\sim)$ is a partition. For every $a \in A$, reflexivity gives $a \in [a]$, so each class is nonempty and $\bigcup_a [a] = A$. Pairwise disjointness of distinct classes is Lemma 1.

Step 2: $\Psi(\mathcal{P})$ is an equivalence relation. Let $\approx := \Psi(\mathcal{P})$.

- Reflexive: each $a \in A$ lies in some $B \in \mathcal{P}$ by (P1), so $a \approx a$.
- Symmetric: $a \approx b$ means $a, b \in B$ for some B ; the same B witnesses $b \approx a$.
- Transitive: if $a, b \in B$ and $b, c \in B'$, then $b \in B \cap B'$, forcing $B = B'$ by (P2); hence $a, c \in B$ and $a \approx c$.

Step 3: $\Phi \circ \Psi = \text{id}$. Let $\mathcal{P}$ be a partition and set $\approx = \Psi(\mathcal{P})$. Pick any $a \in A$ and let $B_a \in \mathcal{P}$ be the (unique) block containing a . The class $[a]_{\approx}$ consists of all x sharing some block with a , but uniqueness of the block makes this exactly B_a . Thus $\Phi(\Psi(\mathcal{P})) = \{B_a : a \in A\} = \mathcal{P}$, since every block of $\mathcal{P}$ contains at least one element.

Step 4: $\Psi \circ \Phi = \text{id}$. Let $\sim$ be an equivalence relation and set $\mathcal{P} = \Phi(\sim)$. For $a, b \in A$,

$$a \Psi(\mathcal{P}) b \iff \exists [c] \in \mathcal{P}, a, b \in [c] \iff c \sim a \wedge c \sim b \iff a \sim b,$$

Read aloud: “ a is $\Psi(\mathcal{P})$ -related to b iff there exists a class $[c]$ in $\mathcal{P}$ containing both a and b , iff c is equivalent to a and c is equivalent to b , iff a is equivalent to b .” using symmetry and transitivity for the last equivalence. So $\Psi(\Phi(\sim)) = \sim$.

The two maps are mutually inverse, hence bijective. $\square$

3 An Example to Hold Onto

Example 1 (Congruence mod n). Fix $n \in \mathbb{Z}_{>0}$. Define $a \equiv b \pmod{n}$ iff $n \mid (a - b)$. This is an equivalence relation on $\mathbb{Z}$, and Theorem 2 shows the residue classes $[0], [1], \dots, [n - 1]$ form a partition of $\mathbb{Z}$ into exactly n blocks.

Example 2 (Divisibility). On $\mathbb{N}_{>0}$, define $a \leq_1 b$ iff $a \mid b$. This is a partial order: reflexive ($a \mid a$), antisymmetric ($a \mid b \wedge b \mid a \Rightarrow a = b$ for positive integers), and transitive. It is *not* total: $2 \nmid 3$ and $3 \nmid 2$.